

IMARTICUS LEARNING

DERIVATIVES – FORWARDS, FUTURES, CFD, FRA & MLN



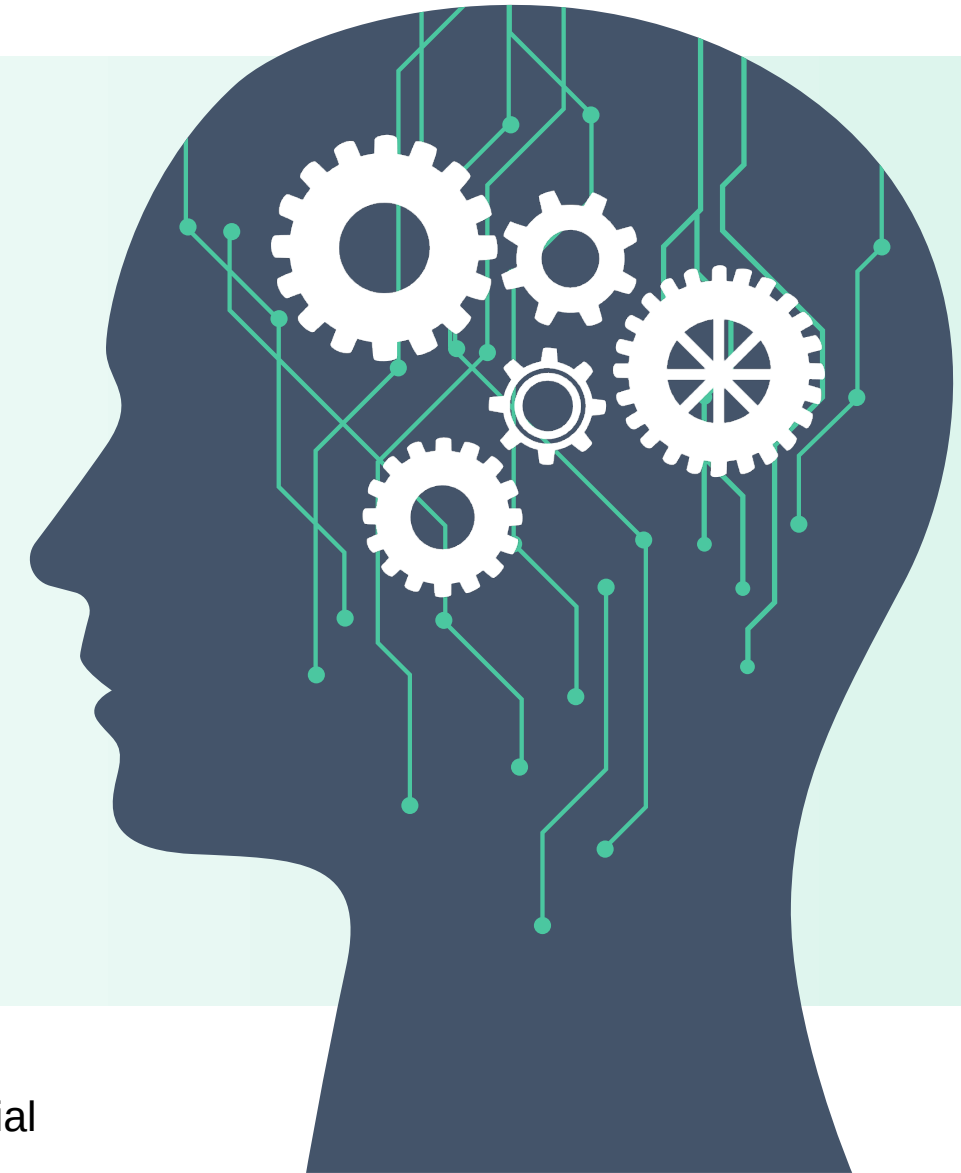
IMARTICUS LEARNING

DISCLAIMER

The training content and delivery of this presentation is confidential, and cannot be recorded, or copied and distributed to any third party, without the written consent of Imarticus Learning Pvt. Ltd.

YOU ARE GOING TO LEARN…

- Derivatives – An Introduction
- Purpose of Derivatives
- Forwards Contracts & its Features
- Futures Contract & its Features
- Futures Margins, Settlement & Payoffs
- CFDs & MLNs
- Forward Rate Agreement



INTRODUCTION TO DERIVATIVES

Derivatives are financial contracts whose values are derived from the values of underlying assets. The underlying of a derivative can be financial assets like equity , bond , interest rate or can be a non financial assets like gold , commodity etc. or even another derivative.

Feature of Derivative Markets:

01 Largest & most liquid market in the world with market size of close to 700 Trillion USD

02 Derivatives have the characteristic of Leverage or Gearing. With a small initial outlay of funds participants can trade big volumes.

03 Derivatives involve complex pricing and are priced using most complex models.

04 Users of derivatives include hedgers, arbitrageurs, speculators and margin traders.

05 Derivatives are traded over-the-counter bilaterally between two counterparties but are also traded on exchanges.

DERIVATIVES: UNDERLYING ASSET

The underlying of a Derivative is an Asset like Shares, Bonds, Commodities, FX etc or Basket of Assets or Index, such that the cash flows of the derivative depend on the value of this underlying.

Underlying Asset Types

Financial Instruments



Equities, Bonds, FX, Loans etc.

Non-financial Instruments



Commodities, Live Stock, Weather, etc.

SPOT PRICE

A derivative's notional value is directly related to the spot price of the security as the Derivatives Contracts valuation depends on the underlying Asset Spot Price.

Spot Price

The Current Market Price of an Underlying Asset in the Cash or Spot Market

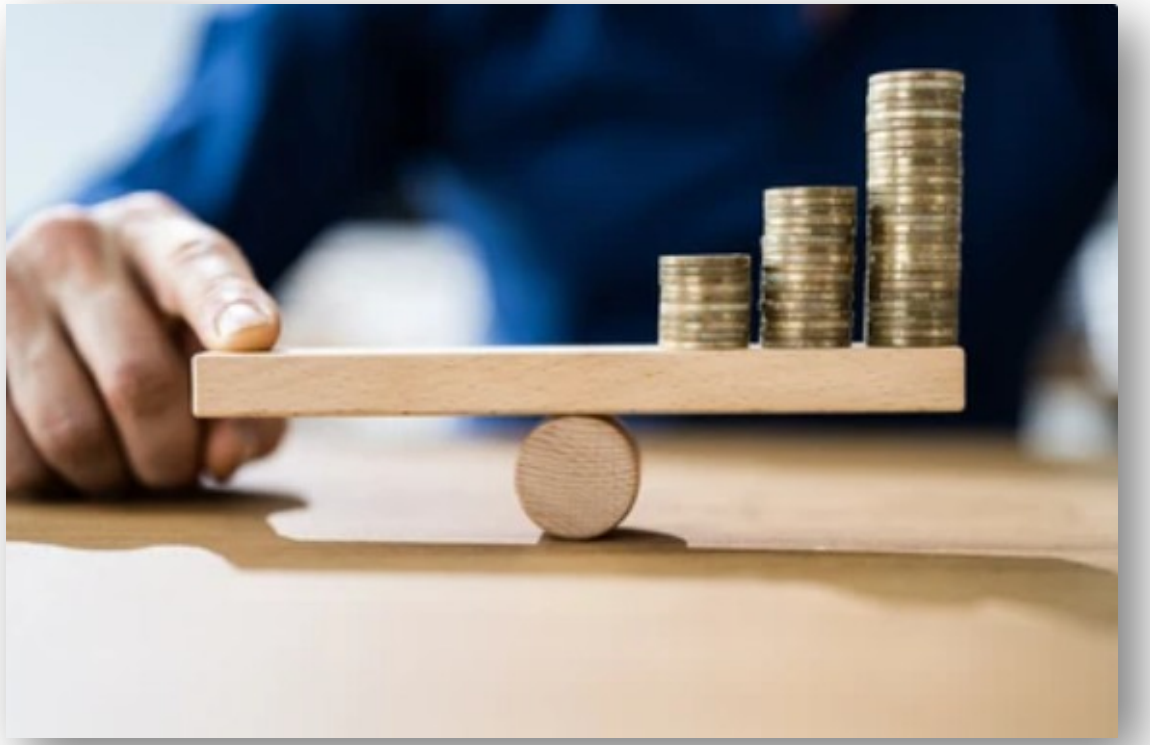
Notional Value

The value of a Derivative Contract underlying asset at the contract price

LEVERAGE

Derivatives need less investment amount just equal to fraction of trade value i.e., Margins or Premiums to take large trade exposure positions in the underlying.

Less Capital is needed than Cash/Spot Market Trades like delivery-based buying of Bonds or Equities etc



LEVERAGE

Derivatives often provide leverage, which means that a relatively small investment (the initial margin or premium) can control a much larger position in the underlying asset. This leverage can amplify both gains and losses.



NOTIONAL VALUE

The total value of the assets in a leveraged position

⬇

Contract size x Agreed Derivative price



x 1000 barrels

Price Agreed = $80

Notional value = $80,000

NOTIONAL VALUE

IMARTICUS LEARNING

Notional value, also known as nominal value, is the total dollar value assigned to a derivative contract, such as a futures contract or options or an interest rate swap. Notional value is used to calculate the amount of cash flow exchanged between the parties involved in the contract.

Formula:

NOTIONAL VALUE = NO. OF CONTRACTS X LOT SIZE (CONTRACT LOT) X CONTRACT PRICE

NOTIONAL VALUE = 10 Contracts X 250 Lot size X $25 Price = $62,500

CONTRACT LOT OR CONTRACT SIZE

The number/quantity of underlying assets contained in one derivative instrument specified by the Exchange.

Purpose:
Standardizing the Quantity in each Derivative contract so as to make them fungible and enable the Trading in Exchange Markets



SETTLEMENT DATE

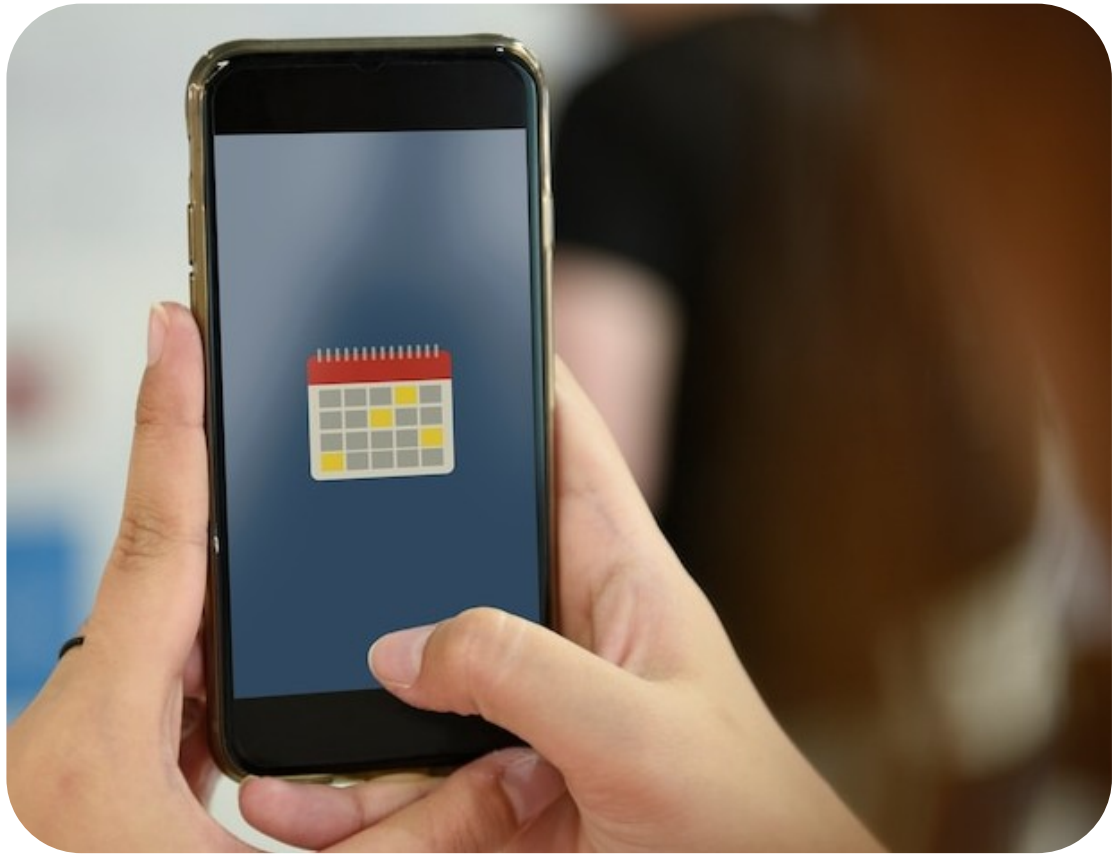
It is the future agreed date wherein the contract obligations are fulfilled and is also called as Delivery Date or Settlement Date or Value Date.

The **Settlement Date/Value Date** in derivatives refers to the date on which the contractual obligations of a derivative contract are settled. It's the day when the actual exchange of cash or the underlying asset occurs between the parties involved in the contract. The specific procedures for settlement can vary depending on the type of derivative and the market in which it is traded.



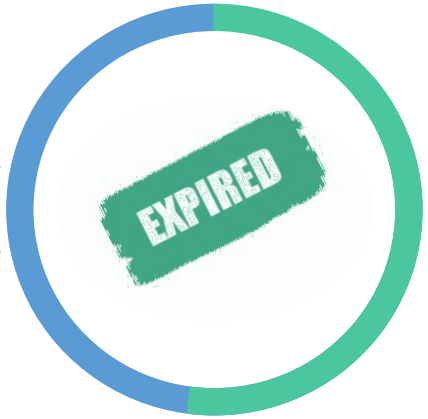
EXPIRY DATE

The expiry date in derivatives refers to the date when a derivative contract comes to an end or reaches its maturity. It's a crucial date because it signifies the point at which the contractual rights and obligations of the parties involved in the derivative cease to exist. The specific implications of the expiry date depend on the type of derivative contract.



EXPIRY DATE

The last day that an Derivative Contracts like Options or Futures Contract is valid



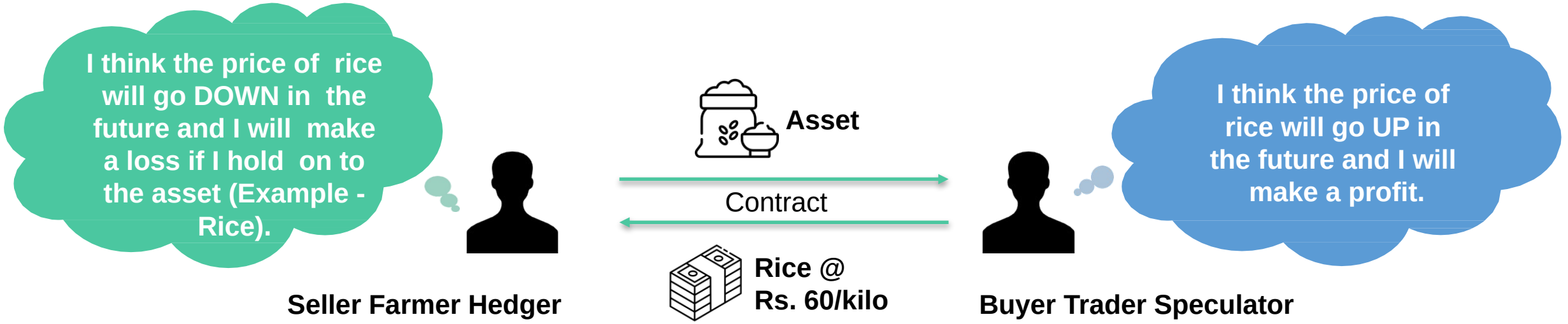
After the Expiry Date all the Derivative Contracts like Options expires and becomes worthless

For all listed stock options in the United States



DERIVATIVES APPLICATION – HEDGING & SPECULATION

Derivatives are also used for speculating and hedging the risks associated with fluctuating prices of assets.



HEDGING

Derivatives can used to hedge if market moves exactly opposite to their expectation

SPECULATION

Derivatives can be used by participants to speculate and to make a profit if the market moves as expected by them.

HEDGING

Scenario: Farmer cultivating Wheat has an exposure to Wheat Price fluctuations risk and wants to protect income from fluctuations in wheat prices. Farmer can achieve this through a forward contract.

Step 1: Protecting Income (Hedging)



Hedging Objective: Farmer want to hedge against the risk of falling wheat prices because a price decline would reduce Farmer's income when eventually sell his wheat crop.



Enter into a Forward Contract (Hedging): Farmer contact a grain buyer or a commodities dealer to enter into a forward contract for the sale of wheat crop. The terms of the contract are as follows:

- Forward Contract Expiration Date: Six months from today
- Forward Contract Delivery Date: Same as the expiration date
- Forward Contract Price (Agreed Upon): $5 per bushel
- Quantity: 1,000 bushels of wheat (representing a portion of expected harvest)



Hedging Outcome: If wheat prices fall over the next six months, Farmer is protected because he has locked in the selling price of $5 per bushel in Forward Contract. Farmer can sell wheat at this price, even if market prices are lower, ensuring a predictable income.

SPECULATION - PROFITING FROM PRICE INCREASE

1. Speculation Objective: Traders who believe that wheat prices may rise due to potential supply shortages. Traders want to profit from this expected price increase.

Enter into a Forward Contract (Speculation): Trader enters into the Forward Contract as a Buyer of Forward Contract with the Seller of Forward Contract for Quantity of 1,000 bushels of wheat. The terms of this contract are as follows:

- Forward Contract Expiration Date: Six months from today
- Forward Contract Delivery Date: Same as the expiration date
- Forward Contract Price (Agreed Upon): $5.50 per bushel
- Quantity: 1,000 bushels of wheat

Speculation Outcome: If wheat prices rise as anticipated over the next six months, the market price of wheat may be higher than $5.50 per bushel, for example say market price is $6.50 per bushel at the settlement date. Trade can take delivery of the Wheat in Forward Contract at Contract Price of $5.50 per bushel and can sell the 1,000 bushels of wheat at the higher market price $6.50, realizing a profit $1 per bushel, making total profit of $1000

EXAMPLE OF SPECULATION

Anyone who buys or shorts a security with the expectation of a favorable price change is a speculator

Company HUL Ltd makes good daily necessities like toothpaste etc. which will always be required. Even in recession, I think this stock will go up! Let me take Long Position in HUL Company Equity Derivatives



SPECULATOR - Bullish View

Company L&T shares will probably decline if the economy starts to deteriorate. I better take Short Position in L&T Equity Derivatives to make profits from the fall in the Prices



SPECULATOR - Bearish View

TYPES OF DERIVATIVES

IMARTICUS LEARNING

Based on Place of Trade



Exchange Traded
Standardized contracts
Market Risk



Over-the-counter
Private, Tailor-made contracts
Market Risk + Counterparty Risk + Liquidity Risk

EXCHANGE TRADED DERIVATIVES

An exchange traded derivative is a financial instrument that trades on a regulated exchange and whose value is based on the value of another asset. Simply put, these are derivatives that are traded in a regulated centralized transparent markets.



No direct contact between seller & buyer

THE WORLD'S LARGEST DERIVATIVE EXCHANGES

IMARTICUS LEARNING

KRX KOREA EXCHANGE

eurex

CME Group

Chicago Board of Trade

NYMEX NEW YORK MERCANTILE EXCHANGE

OTC DERIVATIVES

Contracts that are traded (and privately negotiated) directly between two parties, without going through an exchange or other intermediary.







Long Position - Buyer of the Contract

Short Position - Seller of the Contract

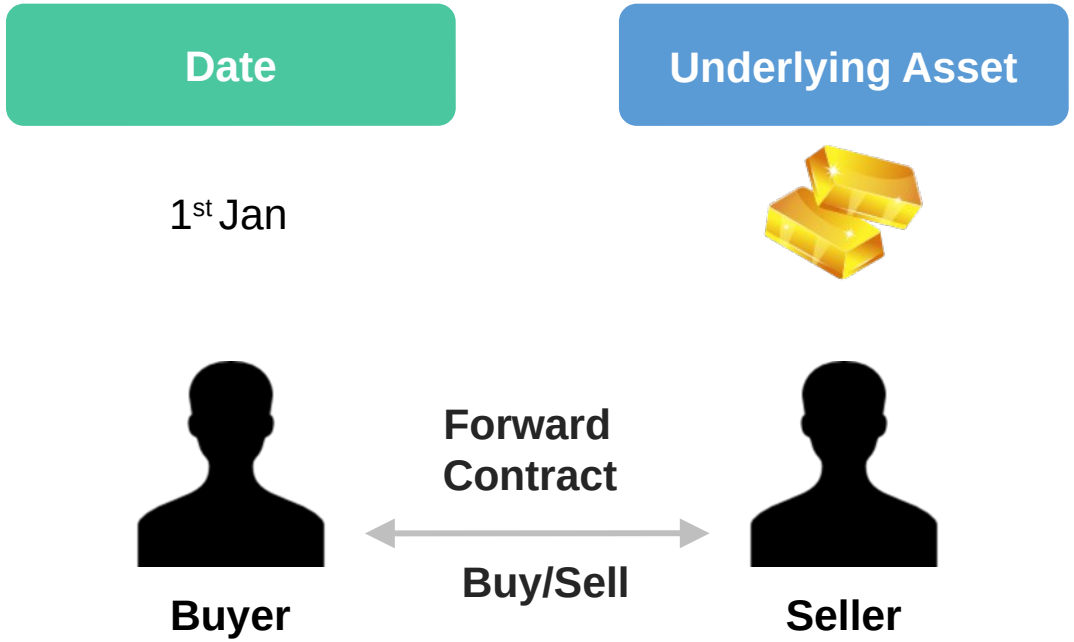
The contracts that are entered in the OTC Market are known as **Bilateral Agreements**.



Forwards Contract

INTRODUCTION TO FORWARDS

Forwards are OTC Derivatives Contract is an agreement to buy or sell the underlying asset on a specified future date at the price, quantity, quality agreed on trade date but performance or settlement of the contract settled on value / settlement date.



No money or product is exchanged on Trade Date as the contract is settled at a agreed future date.....but some Forward Contracts may require collateral/margins to trade in OTC-Dealers Markets

FORWARDS CONTRACT ON SETTLEMENT

Forward Contract is an OTC Derivative Contract wherein the contract is settled at agreed future date

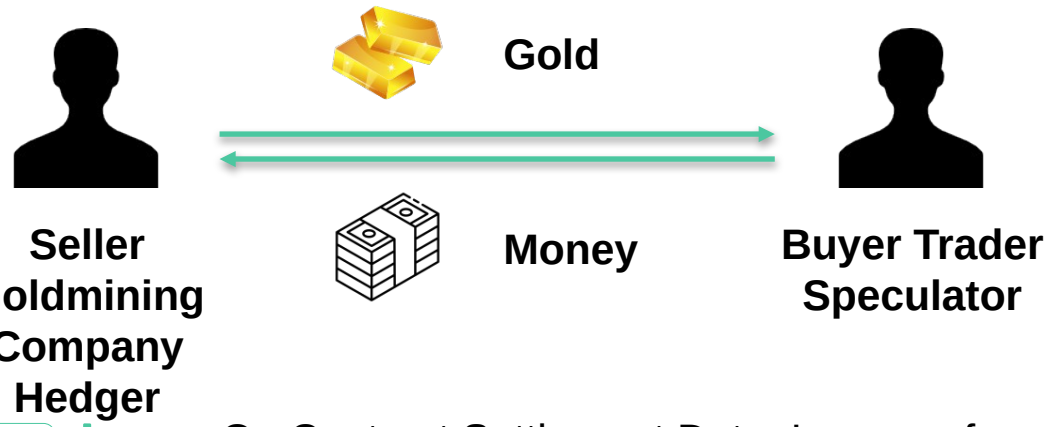
Contract Trade Date

1st June

No payment/Underlying Asset delivery takes place till the expiry date/maturity date of the contract but may involve collateral margins deposits as per the contract terms.

Contract Settlement

31st Dec



On Contract Settlement Date, In case of Physical Settlement, Buyers Pays the Cash and Seller Delivers the Underlying Asset.

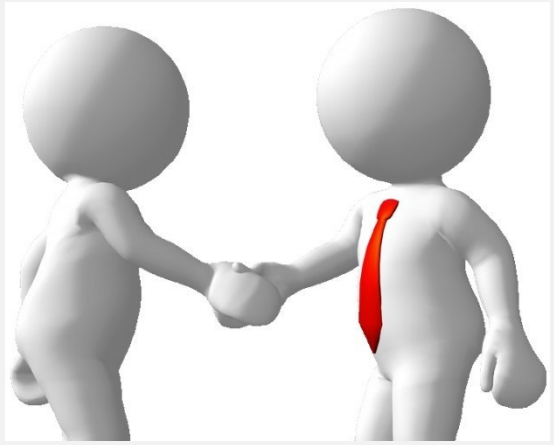
FEATURES OF FORWARDS

1



OTC Contract

2



Independent Agreement

3



Customized

FEATURES OF FORWARDS

4



Collaterals posting depend on the Contract Terms

5



Potential Risk of Default

6



Dependent on Spot Price

FORWARDS CONTRACTS - FUNDAMENTALS

Forward Price

The agreed upon price in a forward contract in called the forward price.

Spot Price

The price quoted in Cash / Spot Market where trades are settled in short span of time like T+1 or T+2 Days.



Settlement Date/ Value Date

The date on which the contract is settled betweeen the counter parties.

RISK IN FORWARDS

IMARTICUS LEARNING

Liquidity Risk	Credit Risk	Less Regulation
		
Difficult to Resell or Exit from the contract	Default Risk faced by the Counter Parties Delivery/Settlement is unsure	Less regulations in comparison to Futures

FORWARD - EXAMPLE



Suppose a Buyer (KKR Hedge Fund) enters into a Forward Contract with a Seller (JPMC) to purchase 1000 Apple Computers Shares at $350 per share. This is a three months contract settled in Physical Settlement. So, after three months, on the date of settlement, the Buyer (KKR Hedge Fund) will have to pay $ 350,000 and the Seller(JPMC) will have to deliver 1000 Shares of Apple Computers Shares.



There can be two cases i.e. either the price of Apple Computer Shares will rise or it will fall in the future. If the price rises, say $355 per share on the settlement date, the buyer will be in a profitable position since Buyeer has already bought at $350. On the other hand, the seller has made a loss of $ 5 per share, since Seller has agreed to sell at the fixed rate of $ 350 while now, Seller could had sold at $355, for instance. While if the price falls, say $340 per share the seller will be in a profitable position, since Seller has already sold at $350 in Forward Contract and the Buyer will be losing position, while the Buyer would had been able to buy at $340 from the market if he had not been in the contract.

FORWARD PRICING

Forward Price Formula

$$F_0(T) = S_0(1+r)^T$$

Where:
F0 = The contract's forward price
S0 = The underlying asset's current spot price
r = The risk-free interest rate that applies to the life of the forward contract
T = The delivery date in years

Example:

An Investment Banking is providing a quote to enter into a 3 month forward contract with one of its client for an investment asset currently trading at $1000. The risk-free rate in the country is 5%. The forward price which the IB can quote minimum of :

F = $10000 *(1+0.05)^3/12
F = $10000 *(1.05)^0.25
F = $1012.27

FORWARD PAYOUT FORMULA

Forward Contracts can be settled using Cash (Profit & Loss) Settlement wherein the Losing party pays Cash (the difference between the prevailing spot price & forward contract price) to the Gaining Party.

Long Position in Forward Contract i.e Forward Contract Buyer
Spot Price (on settlement date) – Forward Price

Short Position in Forward Contract i.e Forward Contract Seller
Forward Price – Spot Price (on settlement date)

FORWARD TRADE ECONOMICS

IMARTICUS LEARNING

- Counterparty 1(Client): KKR Hedge Fund
- Counter Party 2 (IB): Morgan Stanley
- Direction: Buy
- Trade Date: 1st Sept 20XX
- Underlying Asset: Apple Common Equity Shares
- Quantity: 50,000
- Forward Price per Unit/share: $178.00
- Underlying Shares Security Identifier : ISIN : US0378331005
- Settlement Date: 31st Dec 20XX
- Settlement Amount: 50,000 x $178.00 = USD 8,900,000



Futures Contract

FUTURES CONTRACT

Futures is an Exchange Traded Derivative Contract and it is a legally binding agreement to buy or sell a commodity or financial instrument in a designated future month at a price agreed upon at the initiation of the contract by the buyer and seller. Futures contracts are standardized according to the quality, quantity, and delivery time and location for each commodity.

Both Parties have Obligations



Durations specified by exchanges Futures are also known as **Exchange Traded Forward contracts**.

FUTURE CONTRACT STANDARDIZATION



Settlement Date
Last Thursday or 3rd Friday of the month (depending on the market)



Contract Value
The contract value will be the same for both buyers and sellers.



Lot Size
Lots refer to the no. of the underliers in each contract. If the lot size is 500, then each contract will derive it value from 500 underliers.



Price Tick
The price movement of a future is also standardized. If the price tick is Rs 0.50, then the price can move from Rs 100 to 100.50 to 101.



Margins
The margins are pre-determined by the Exchange/ Clearing House and hence is standardized across all the participants in
the market.



Quality
Incase of commodities, the quality of the underlier is standardized – in terms of 22 karat or 24 karat or 18 karat Gold



Currency Pairs
In the case of currency futures, standard pairs of currencies are used. For e.g. in the NSE: only USDINR, EURINR, GBPINR and JPYINR can be traded

LONG AND SHORT POSITION IN FUTURES CONTRACT



Trader A Long Position Bought Futures

Price ↑

Profit ↑



Trader B Short Position Short Futures

Price ↓

Profit ↑

FUTURES

IMARTICUS LEARNING

1. The buyer of the contract is said to be long and the party selling the contract is said to be '**short**'.
2. Future contracts (FUTURES) are traded on an Recognized Exchanges with the help of Members or Brokers of the Exchange and the Counter Parties are not known to each other.
3. Given that futures are standardized contracts, the contract size, quality, duration of futures contracts are specified by the exchanges.
4. Future contracts trading requires Collateral Margins as specified by the Exchange/Clearing House

FUTURES CONTRACT: EXAMPLE

IMARTICUS LEARNING

Sell Higher | **Buy Lower**

I think the price of rice will go DOWN in the future and I will make a loss if I hold on to the asset.



Seller



Asset

Contract





Buyer

I think the price of rice will go UP in the future and I will make a profit.

Cash price of rice

Short-sale of the futures contract

FUTURES CONTRACT - INITIAL MARGIN

To mitigate the counterparty, risk the exchange/ clearing house charges various margins:

Initial Margin:

The amount paid at the start of the contract is known as Initial margin. This is the amount paid by both the buyer & seller of the contract and is required to initiate the trade.

The initial margin consist of SPAN (standardized portfolio analysis of risk) margin & exposure margin. The SPAN margin is calculated using Value at Risk(VAR) method and is based on the liquidity & the volatility of the stock. The SPAN margin is mandatory .

The exposure margin is an additional margin which is optional and is collected as an additional buffer of safety.
Ex: Stock Price is 1820 & lot size is 400. The initial margin is 17 % (SPAN – 11 & Exposure margin – 6 %)
IM = 728000 (TCV) * 17 % = 123,760

MAINTENANCE MARGIN & MTM MARGIN

Maintenance Margin is the minimum amount of margin balance that need to maintained in the account to maintain a futures position. Any decrease below the maintenance margin level will trigger "**Margin Call Amount**" which the Trades must pay to their Brokers/Members to keep their Futures Position else their positions will be closed/square off due to failure of paying the Margin Call Amount.

Mark to Market (MTM) or Variable Margin:

All Futures Positions are daily valued at the Closing Price of the Futures Contract & daily Mark - to- Market Margins are debited/credited to the Traders Account by the Brokers/Members of the Exchange and this is also known as Profit /Loss Margin and settled on Daily Basis. MTM margin is charged by an Exchange/Clearing House when profit or loss arises due to the positions taken by the participants are realised/collected to keep the contracts solvent.

MTM Pnl – Day 1 = (Closing Price C1 – Buying Price)*Lot Size*Number of Lots

MTM Pnl – Day 2 = (Closing Price of Day 2 C2 – Closing Price of Day 1 C1)*Lot Size*Number f Lots

MTM Pnl – Day 3 = (Closing Price of Day 3 C3 – Closing Price of Day 2 C2)*Lot Size*Number of Lots

MTM Pnl – Day 4 = (Selling Price – Closing Price of Day 3 C3)*Lot Size*Number of Lots

FUTURES – CASE STUDY

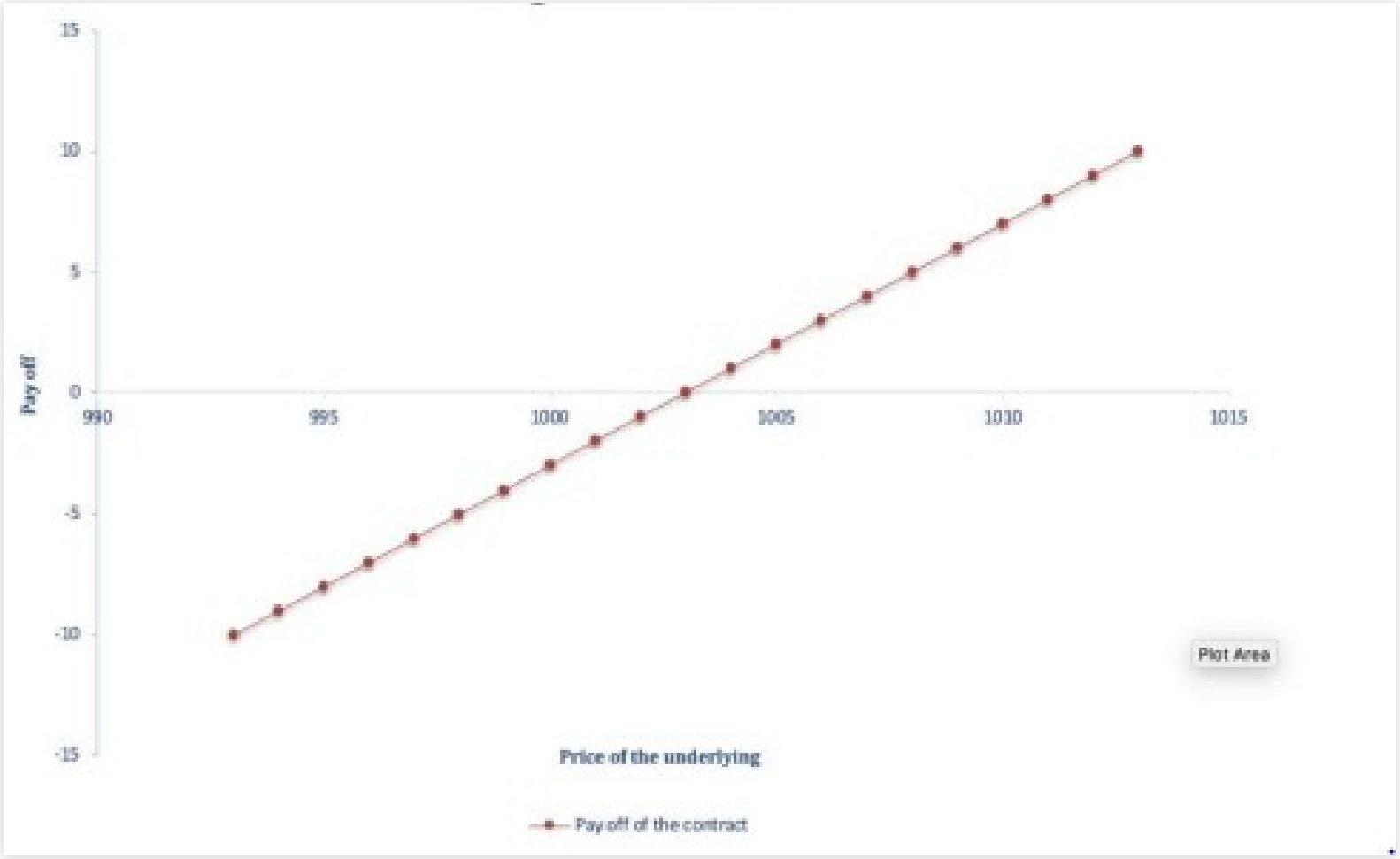
Case Study

Stock Futures Price is 1820$ & lot size is 400. The initial margin is 17 % (SPAN – 11% & Exposure margin – 6 %) and maintenance margin 15 %.
IM = 728000 (TCV) * 17 % = 123,760 $
Maintenance Margin – 728000 * 15 % = 109 ,200$

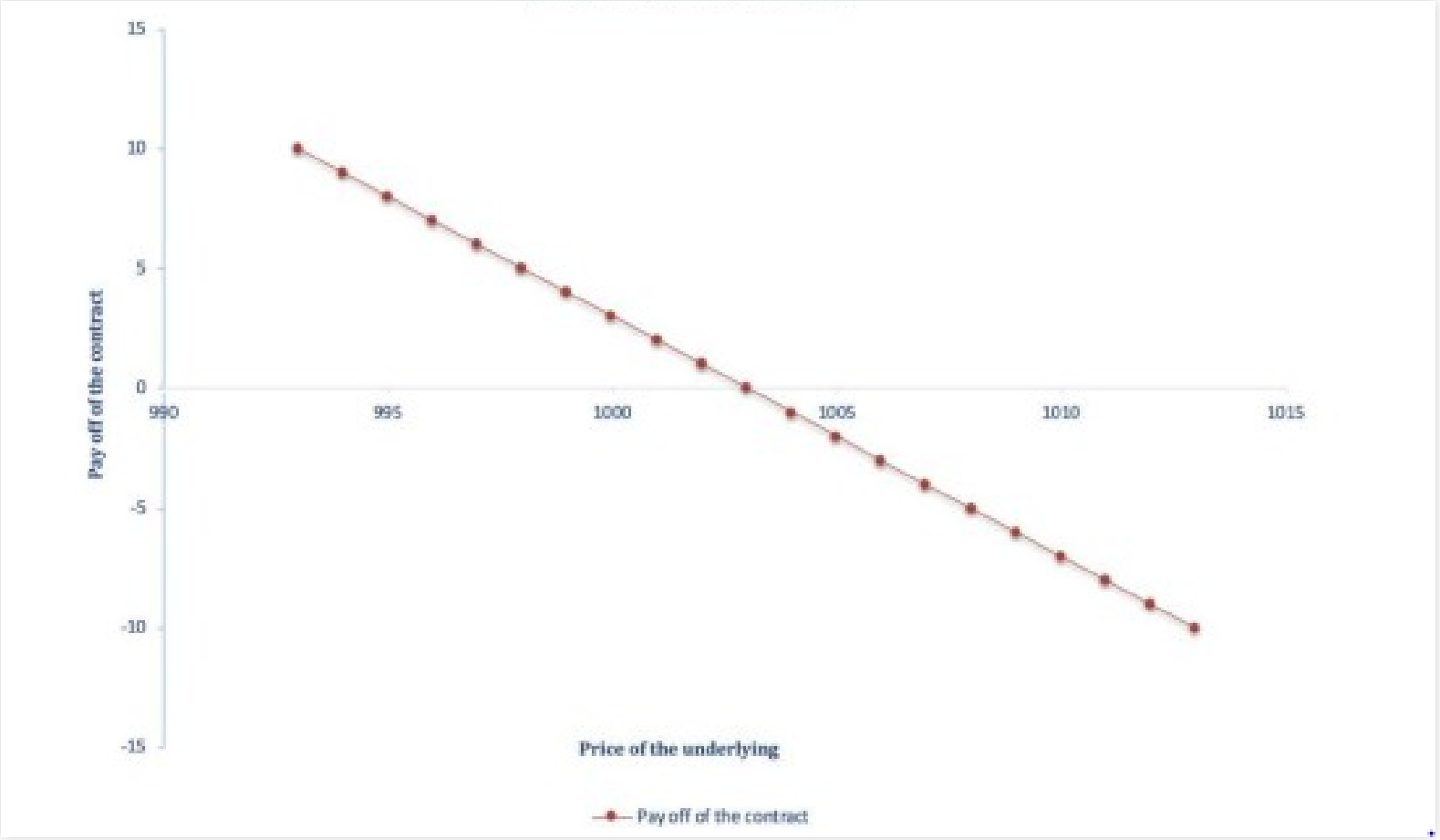
Position Type – Long Position i.e Buy Position on Stock Futures

- Contract Price - 1820 $
- Day-2 Closing Price of Futures – 1800 $
- MTM Loss of 20 * 400 = - 8000 $
- 8000 $ is deducted from 1,23,760 $ & the new balance is – 115, 760 $
- Day-2 Closing Price – 1780 $
- Loss of 20 * 400 = - 8000 $
- 8000 is deducted from 1,15,760$ & the new balance is – 1,07,760$ which is below the minimum margin level. Margin call will be triggered and the trader need to provide 16,000 $ to bring the Margin to the original required level of 123,760 $.

LONG POSITION IN FUTURES (BUY POSITION)
LINEAR PAY OFF : SPOT PRICE – FUTURE PRICE (CONTRACT PRICE)



SHORT POSITION IN FUTURES (SELL POSITION): LINEAR PAY OFF FUTURE PRICE (CONTRACT PRICE) - SPOT PRICE



FUTURES SETTLEMENTS

Futures Settlements

Physical Settlement

The amount specified of the underlying asset of the contract is delivered by the seller of the contract to the exchange and by the exchange to the buyers of the contract. Physical delivery is common with commodities and bonds.

For example, NYMEX (Future Exchange) crude oil futures uses this method upon expiry.

Cash settlement

No Physical delivery of instruments - Profit /Loss settled in cash - Common for Index futures.
A cash payment is made based on the underlying Asset Closing Price on Expiry Date or Reference rate, such as a short-term interest rate index such as Euribor, or the closing value of a stock market index. The parties settle by paying/receiving the loss or gain related to the contract in cash when the contract expires.

For example, ICE Brent Futures uses this method. (The only condition for the usage of these methods is the contract must expire)

FUTURES TRADE ECONOMICS

- Counterparty 1(Client): Alliance Bernstien
- Counter Party 2 (Exchange Broker): JPMC
- Direction: Buy
- Trade Date: 11/09/20XX
- Underlying Asset: Apple Computers Equity Shares
- Quantity: 50,000 Shares
- October Series Apple Futures Price per Unit/share: $182.00
- Underlying Shares CUSIP /ISIN/SEDOL: US0378331005
- Expiry Date : 30/10/20XX

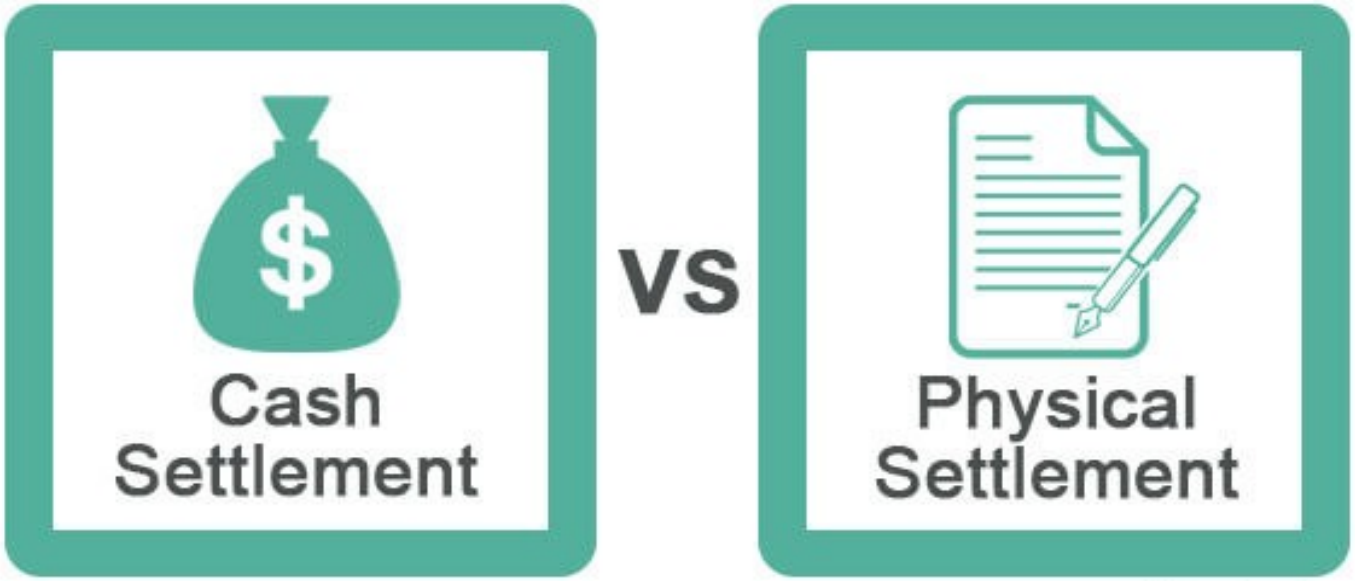
SETTLEMENT MODE IN FUTURES CONTRACT

Cash Settlement

The trade will settle in cash without the exchange of the underlying asset.

Physical Settlement

The trade will settle with the underlying asset being delivered.



Note: The only condition for the usage of these methods is the contract must expire

OPEN INTEREST - OI

Open Interest indicate total number of open contracts (still not closed) for a security /index .It applies for Futures & Options market.

Why OI?

The Open Interest shows the total number of contracts/positions active in the underlying asset for that Futures Series and used to gauge market sentiment, interest & liquidity and also used as a trend indicator.

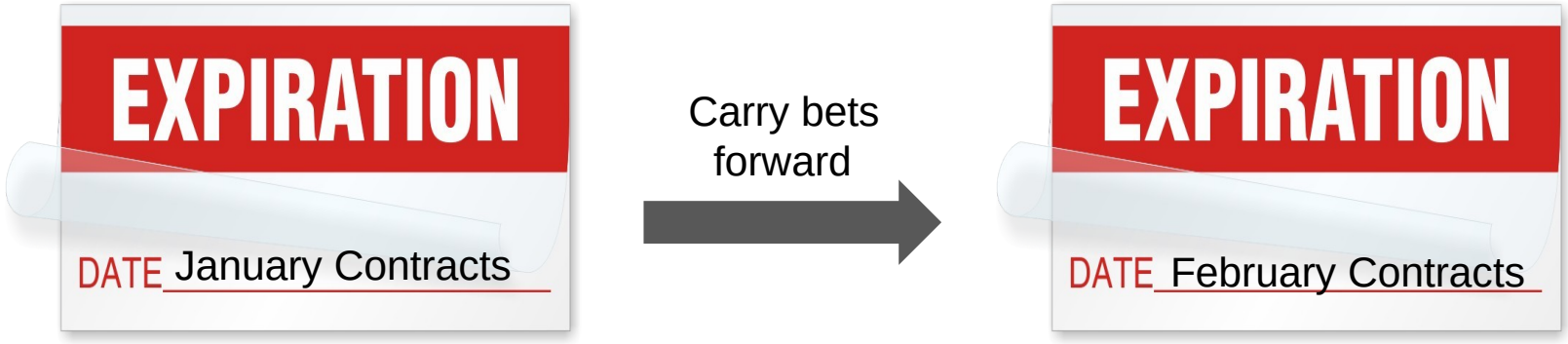
Open Interest Calculation:

Let's assume the market consists of 7 traders A, B ,C D, E, X & Z who trades Reliance Industries (RIL) Stock Futures and each contract lot is 500 shares

- **Monday:** A buys 6 contracts & C buys 4 contracts & D sells all 10 contracts, So the total OI is 10 contracts i.e 5000 (10x500) RIL Shares
- **Tuesday:** D buys 8 contracts and the Seller is E .The Total OI will still remain as 10 (5000 RIL Shares) as 8 contracts are mere transfers as D reduces his position but E increases his position by 8 contracts.
- **Wednesday:** D sells 5 more contracts. **A** buys 3 & C buys 2 .Now the total OI is 15 Contracts (7500 RIL Shares).
- **Thursday:** X enters the market & sells 25 contracts and Z buys these contracts and Now the total OI is 40 i.e 20000 RIL Shares
- **Friday:** A closes his position by selling 9 contracts and X buys these contracts and Now the total OI is 31 i.e 15500 RIL Shares since A & X reduces their positions by 9 contracts

ROLLOVER

A process in which investors carry forward their positions in a derivatives contract from one expiry date to another expiry date by closing current derivative contract and initiating new position in another expiry date derivative contract in the same underlying Asset



In India, equity derivatives expire on the last Thursday of each month

Week before expiry

Till the last minute

Rollovers

DIFFERENCE BETWEEN FUTURES & FORWARDS

IMARTICUS LEARNING

Futures **v/s** **Forwards**

- Exchange Traded Contract
- Standardized contract
- Fungible and hence Liquid
- Daily MTM Settlement
- Margins
- No counter party risk
- Counterparties are Unknown

- Over The Counter Contract
- Customized contract
- Bilateral contract hence Illiquid
- End of the period settlement
- Margins/Collaterals depends on the Contract Terms
- Counter party risk
- Counterparties must Know each other to enter into the Contract

CFD – CONTRACT FOR DIFFERENCE

- The CFD is an agreement (contract) to exchange the difference in value of a particular currency, commodity , share or index between the time at which a contract is opened and the time at which it is closed.
- It is a contract between two parties to speculate on the movement of an asset price.
- The parties entering into a CFD do not need to buy the underlying or do not required have the underlying to short .
- The contract payout will amount to the difference in the price of the asset between the time the contract is opened and the time it is closed.
- If the asset rises in price, the buyer receives cash from the seller, and vice versa.
- CFDs are traded on leverage to give traders more trading power, flexibility and opportunities.

CFD – CONTRACT FOR DIFFERENCE

Example :

Let's say that counterparty bought CFDs at a price of $50
With a margin capital of $10 000 the number of CFDs bought would have been 200, that is 10000/50 = 200. The value of the CFD climbs to $55 at which point the buyer decide to close the position .
The difference in asset value is $55 - $50 = $5, which gives as $50 profit per CFD .The total profit earned is $5 * 200 = $1000 or 10 % of the investment.

CONTRACT FOR DIFFERENCE – TRADE ECONOMICS

- Counterparty 1
- Counterparty 2
- Buy/Sell
- Underlying Asset Product Code
- Trade Date
- No of Contracts
- Initial Margin on Current Position
- Life of Trade (In Days) – Max number of days a trade can be open

MARKET-LINKED NOTES

Market-Linked Notes

Market-Linked Notes are investment product that combines a fixed-income investment with additional potential returns that are tied to the performance of equities. Market linked notes are usually structured to return the initial investment with a variable interest portion that depends on the performance of the linked asset.

At maturity, if the underlier increases in value, investors will receive their principal plus a supplemental redemption amount, subject to the maximum payment. If the underlier declines or does not appreciate in value, investors will receive their initial investment.

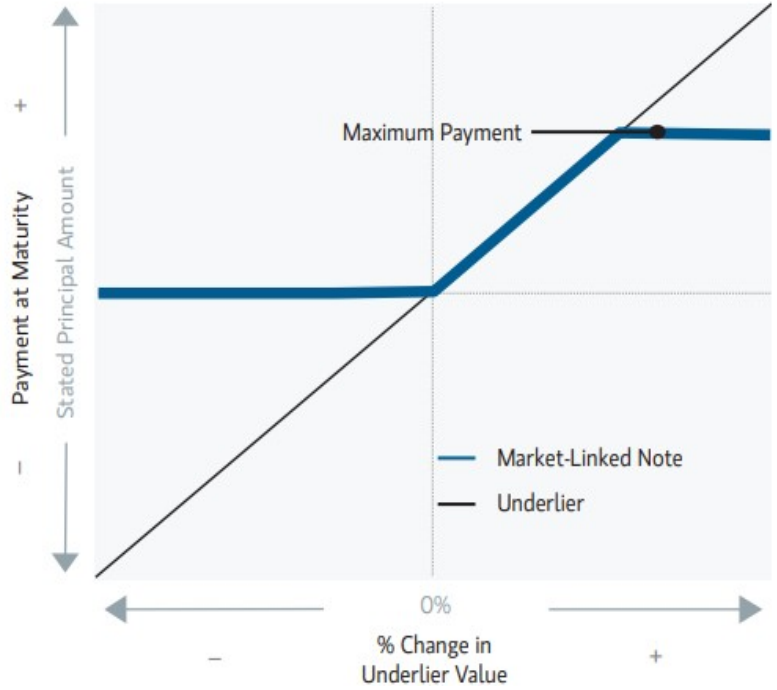
Investor Profile Market-Linked Notes may be appropriate for investors who are concerned about principal risk but seek a return based on the underlier and who are willing to forgo current income in exchange for the repayment of principal at maturity plus the potential to receive a supplemental redemption amount.

The Market-Linked Notes will not be listed on any securities exchange and have limited secondary market.
The estimated value of the securities is determined by reference to pricing and valuation models of the issuer or an affiliate of the issuer, or calculation agent.

Examples of MLN: Equity Linked Notes

MARKET-LINKED NOTES PAY OFF

Hypothetical Payoff Profile



Hypothetical Returns at Maturity

Underlier Price Return	Payment at Maturity	Market-Linked Note Return
+80%	Maximum payment at maturity is reached, note underperforms direct investment in the underlier	+50%
+30%	100% upside participation in underlier performance	+30%
-30%	Note is redeemed for stated principal amount	0%

MARKET LINKED NOTE TRADE EXAMPLE

Let's assume that an issuer issues MLN assuring maximum payment of 150 % of principal & minimum payment of 100 % of the principal. The MLN issued is a fixed income instrument and the return are linked to another asset in the market .However the fixed income instrument will not pay any coupon or interest

If the underlying instrument earns a return of say 80 %, the investor will get maximum payment of 150 % of the investment at maturity in this case.

If the underlying instrument earns a return of say 30 %, the investor will get 130 % of the investment at maturity in this case.

If the underlying instrument earns a return of say - 25 %, the investor will get only the principal that was invested .

Hypothetical Terms

Maturity	7 Years
Participation Rate	100%
Maximum Payment at Maturity	150% of Principal[1]
Minimum Payment at Maturity	100% of Principal[2]
Interest	None

[1]Applicable only if the final value of the underlier is greater than the initial value of the underlier
[2]All payments are subject to the credit risk of the issuer (and the guarantor, if applicable).

FORWARD RATE AGREEMENT



A forward rate agreements (FRA) is a Cash Settled OTC contract between parties that determines the rate of interest to be paid on an agreed-upon date in the future. It is an agreement to exchange an interest rate commitment on a notional amount.



The forward rate agreement determines the rates to be used along with the termination date and notional value. FRAs are cash-settled. The payment is based on the net difference between the interest rate of the contract and the floating rate in the market—the reference rate. The notional amount is not exchanged.

Formula and Calculation for a Forward Rate Agreement

$$\text{FRAP} = \left(\frac{(R - \text{FRA}) \times NP \times P}{Y}\right) \times \left(\frac{1}{1 + R \times \left(\frac{P}{Y}\right)}\right)$$

where:

$\text{FRAP} = \text{FRA payment}$

$\text{FRA} = \text{Forward rate agreement rate, or fixed interest rate that will be paid}$

$R = \text{Reference, or floating interest rate used in the contract}$

$NP = \text{Notional principal, or amount of the loan that interest is applied to}$

$P = \text{Period, or number of days in the contract period}$

$Y = \text{Number of days in the year based on the correct day-count convention for the contract}$

FRA - Case Study

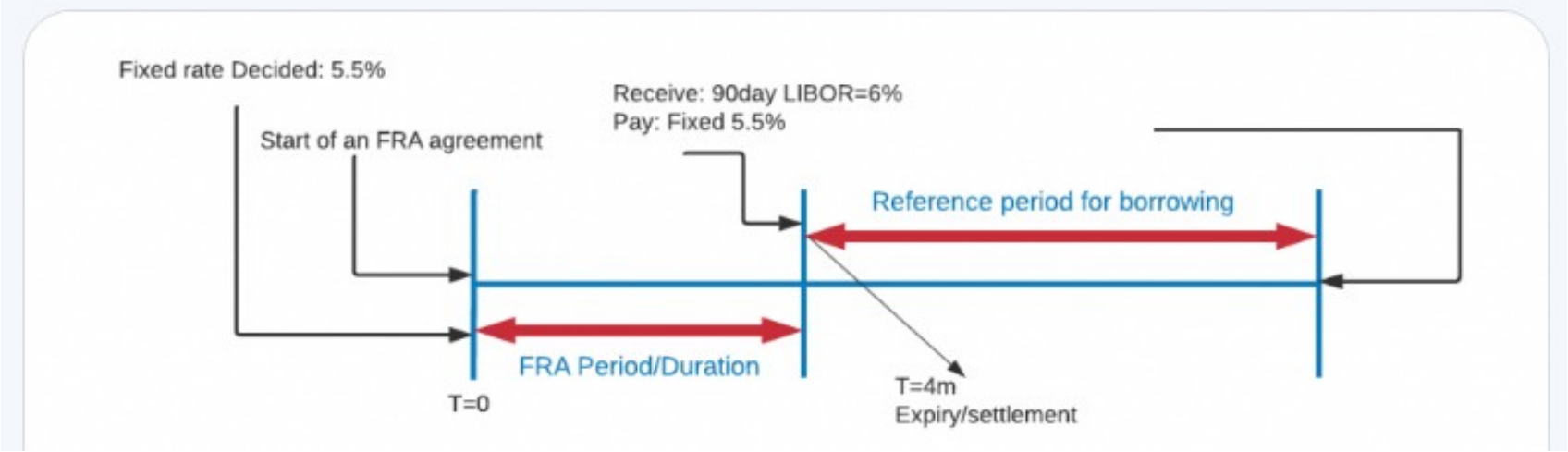
Example 2: FRA Valuation

Suppose we have a 1 x 4 FRA with a notional principal of $1 million. At contract expiration, the 90-day LIBOR at settlement is 6% and the contract rate is 5.5%.

The value of the FRA at maturity is closest to:

Solution

$$\text{Payment to the Long} = \text{Notional principal} \times \frac{(\text{Rate at settlement} - \text{FRA rate}) \times \frac{\text{Days}}{360}}{1 + \text{Rate at settlement} \times \frac{\text{Days}}{360}}$$



FRA - Case study

Since the settlement rate is higher (6%) than the contract rate (5.5%), the buyer will be receiving money from the seller. The payment to the long at settlement are as follow:

Interest saving:

$$= (6\% - 5.5\%) \times \frac{90}{360} \times 1\text{m} = 1,250$$

Discounted back 90 days @ 6%:

$$= \frac{1,250}{[1 + \frac{90}{360} \times 6\%]} = 1,231.53$$

FORWARD RATE AGREEMENT - TRADE ECONOMIES

IMARTICUS LEARNING

1

General Terms: Counter Party A, Counter Party B, Notional Amount, Trade Date, Effective Date, Termination Date, Period End Date, Termination Date adjustment for Unscheduled Holiday: Calculation Agent

2

Fixed Rate Payer:
Fixed Rate Payer: [Counter Party A], Fixed Rate Payer Currency, Payment Dates, Period End Dates, Fixed Rate Day Count Fraction

3

Floating Rate :
Floating Rate Payer: [Counter Party B], Floating Rate Payer Currency, Payment Dates, Floating Rate Reference Benchmark, Floating Rate Reset Dates, Floating Rate Day Count Fraction, : FRA Discounting Rate, Discounting Day Count Fraction.



IMARTICUS LEARNING

THANK YOU !!!

Contact us today!
https://imarticus.org/